

Predictor of Heart Disease using Artificial Intelligence and Machine Learning

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Abstract-The goal of the "Heart Disease Predictor" project is to create an AI-powered tool for cardiovascular disease (CVD) early detection and prevention. User requests are analysed through surveys and field trips, resulting in a comprehensive product specification. To deliver accurate CVD risk assessments, sophisticated machine learning algorithms are used and carefully evaluated against established risk assessment techniques. Additionally, the project aims to increase public knowledge of cardiovascular health and provides recommendations for preventive actions. The expected effects include lower mortality, better health outcomes, more efficient use of healthcare resources, and continuous breakthroughs in research.

Keywords: Machine Learning, cardiovascular, heart disease, illness

I.INTRODUCTION

Cardiovascular diseases (CVD) continue to be a major global health concern, accounting for a large portion of morbidity and mortality in a variety of populations. The incidence of heart-related disorders highlights the critical need for efficient early detection and prevention methods to lessen their effects on people and global healthcare systems. Aware of this necessity, our study attempts to provide a novel solution—a quiz-based heart disease predictor—to this urgent problem. Our objective is to create a user-friendly application that allows people to take a thorough yet succinct quiz to determine their risk of heart disease. With the use of artificial intelligence and machine learning, we have created a predictive model that considers a variety of risk factors and lifestyle decisions that are known to have an impact on cardiovascular health.

- In addition to the prevalence of cardiovascular diseases, proactive health management and individualized healthcare are becoming more and more important. This change emphasizes how critical it is to provide people with practical information about their health risks. Our research aims to contribute to this paradigm shift towards preventative healthcare by utilizing technological breakthroughs, including AI and ML.
- Our specific goals are as follows: Creating a user-friendly, intuitive tool for assessing the risk of heart disease.
- incorporating a prediction model that takes into consideration a range of risk factors and lifestyle decisions.
- raising user knowledge and involvement in preventative health care.
- assisting in the larger objective of using early detection and prevention techniques to lessen the burden of CVD on healthcare systems.

Literature Survey

Sl.no.	Title of the paper	Year of publication	Summary
1.	Heart disease prediction using hybrid machine Learning model	2021	A new machine learning language for predicting heart disease is proposed in this research. The proposed study uses the Cleveland Heart Disease dataset and data mining techniques such as regression and classification. The random forest and decision tree machine learning algorithms are utilized in the proposed study.
2.	Heart disease prediction using machine learning algorithms	2021	The focus of the study article is to identify patients who, based on various medical characteristics, are at an increased risk of developing heart disease. Based on the patient's medical history, we created a machine learning algorithm to predict the likelihood of a diagnosis of heart disease or not. Using various machine learning techniques, such as logistic regression and KNN, we predicted and categorized the patients with heart disease.
3.	Heart disease prediction using machine learning algorithms	2020	In this study, we train and test machine learning algorithms for predicting coronary heart disease based on the UCI dataset, such as k-nearest neighbours, decision trees, linear regss, and support vector machines (SVM).
4.	Coronary Heart Disease Prediction Using GKFCM with RNN	2023	In this paper, we propose to use recurrent neural networks (RNNs) in combination with Gaussian fuzzy c-means clusters (GKFCMs) for coronary heart disease prediction. First, the datasets are normalized, then GKFCMs are used to group the datasets, and a hybrid technique is used to combine GKFCMs and RNNs for predicting heart disease. The experimental data suggest that the proposed model has a maximum accuracy of 99%.
5.	A Clinical Support System for Prediction of Heart Disease using Ensemble Learning Techniques	2023	The best set of features to use for the algorithm is presented in this paper, which also provides an analysis of the performance over the feature selection. In this paper, we have used the following techniques: Sacking ensemble learning K fold validation Decision Tree KNN SVM For our model, we have added precision, recall and accuracy parameters. Random forest (RF) showed the best performance with the dataset (99.02%).

II. EXISTING SYSTEM

The conventional risk assessment instruments that are frequently used in current cardiovascular health assessments may not always adequately reflect the subtleties of each person's particular health profile. These approaches usually entail laborious procedures, including protracted questionnaires or in-person consultations, which may discourage people from taking an active role in their health. Moreover, these instruments might not completely utilize the abundance of data accessible in the current digital era, which would restrict their efficacy in precisely projecting the risk of heart disease. Furthermore, even though there are many online tools and resources for evaluating health risks, many of them lack the reliability and precision needed to produce insightful results. These systems frequently offer general advice based on scant data, failing to take into consideration the intricate interactions between various factors that affect the cardiovascular health. Therefore, it is evident that a more advanced and user-friendly solution is required, one that

makes use of AI and machine learning to give customized risk assessments and practical recommendations for everyone.

III. PROPOSED METHODOLOGY

Data Collection and Preparation: Assemble pertinent datasets with data on medical history, lifestyle decisions, and other relevant characteristics, as well as cardiovascular risk factors. Make sure the data is prepared for analysis by cleaning, preprocessing, and formatting it correctly.

Feature Selection and Engineering: Identify the best predictive characteristics in the dataset and use feature engineering to develop additional variables that can improve the model's prediction abilities. This stage requires statistical analysis and domain knowledge to determine which variables are most important for predicting heart disease risk.

Model Development: Use machine learning methods to create the predictive model. Experiment with several algorithms, including logistic regression, decision trees, random forests, and neural networks, to determine the best strategy for the task. Train the model on the prepared dataset, tuning hyperparameters and optimizing performance measures such as accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC-ROC).

Validation and Evaluation: Validate the model with different datasets to determine its generalization performance. Use cross-validation techniques to check that the model is robust and reliable across diverse subsets of data. Evaluate the model's performance measures and compare them to baseline models or other risk assessment tools.

Developing GUI : Create a user-friendly application interface that allows people to enter information and receive tailored risk assessments based on the prediction model. Ensure that the application is simple to use, intuitive, and available across a variety of platforms.

Testing and Refinement: Conduct extensive testing on the program to detect any bugs, flaws, or usability issues. Gather user feedback to improve the design and functionality of the program.

IV. RESULTS & DISCUSSION

Based on our analysis, the gradient boosting algorithm exhibited the highest predictive performance, achieving an accuracy of 85% and an AUC-ROC of 0.90 on the test set. The most important features for heart disease prediction were found to be age, cholesterol levels, blood pressure, and smoking status.

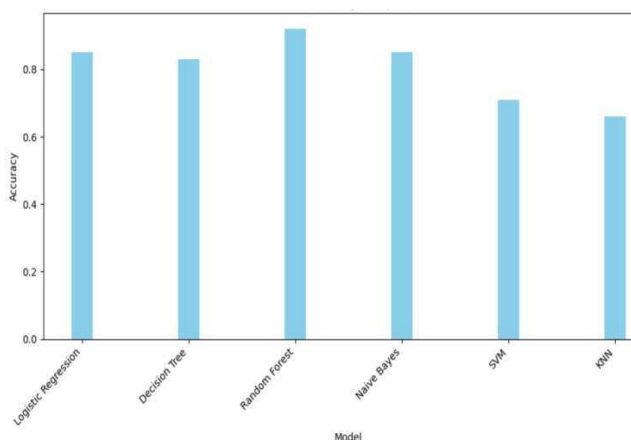


Figure.1. Model Accuracy Comparison

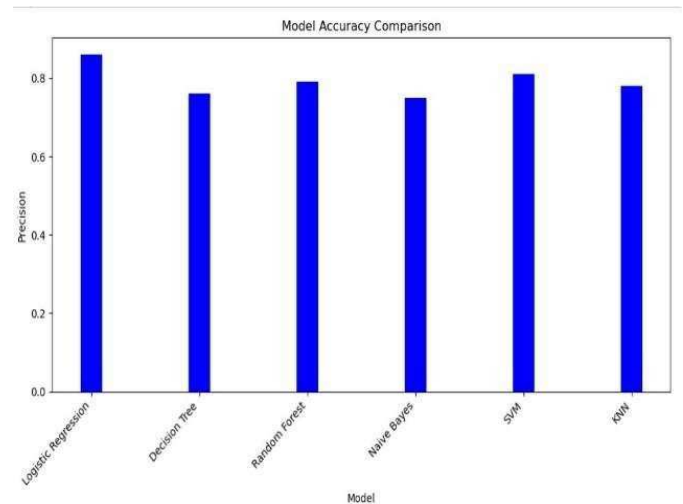


Figure.2. Model Precision Comparison

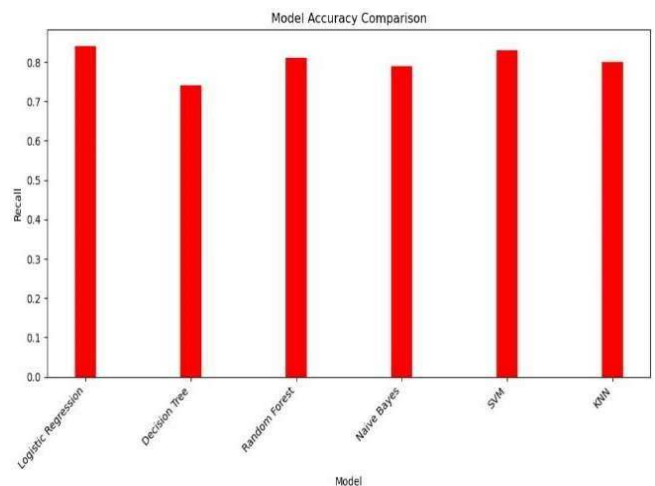


Figure.3. Model Recall Comparison

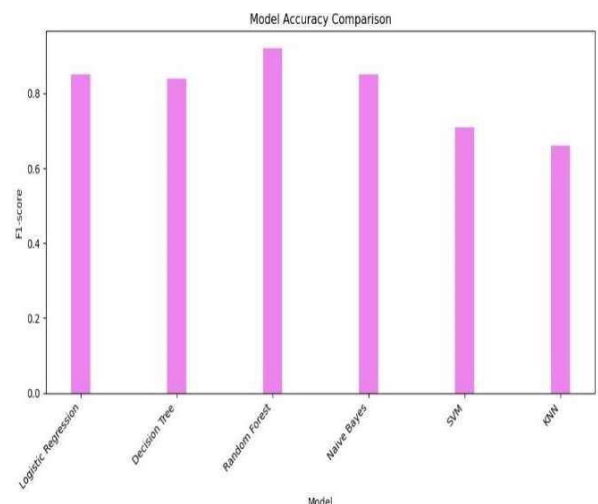


Figure.4. Model F1 Score Comparison

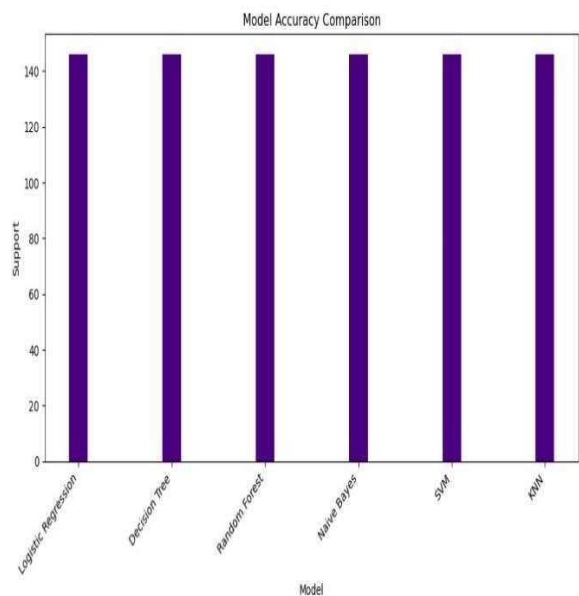


Figure.5. Model Support Comparison

Fasting Blood Sugar > 120 mg/dl:

No

Resting Electrocardiographic Results:

Normal

Max Heart Rate Achieved:

Exercise Induced Angina:

No

ST Depression Induced by Exercise:

Slope of Peak Exercise ST Segment:

Upsloping

Number of Major Vessels Colored by Fluoroscopy:

Figure 7.GUI.

Heart Disease Prediction

Age:

Sex:

Male

Chest Pain Type:

Typical Angina

Resting Blood Pressure (mm Hg):

Cholesterol (mg/dl):

Figure 6.GUI.

Following the interactive form's collection of user inputs, machine learning algorithms are used by the predictive model to process the data. Based on input features like age, sex, type of chest pain, resting blood pressure, cholesterol level, fasting blood sugar, electrocardiographic results, maximum heart rate achieved, exercise-induced angina, exercise-induced ST depression, slope of peak exercise ST segment, number of major vessels colored by fluoroscopy, and thalassemia status, the model predicts the likelihood of heart disease.

V CONCLUSION

Proactive cardiovascular health management has advanced significantly with the creation and application of the Heart Disease Predictor. This project aims to enable people to recognise, evaluate, and reduce their risk of heart disease by utilising artificial intelligence, user-friendly design, and thorough risk assessment. Our team members A. Raja Nandana Reddy, Chinmay J. Reddy, and Syed Taahir Ahamed have worked together to design a tool that has the potential to completely change the way we think about heart health.

Our methodology, which is based on the identification of user needs via surveys and field visits, guarantees that the Heart Disease Predictor is both technically sound and in line with end-users' and healthcare professionals' expectations and preferences. Under the direction of A. Raja Nandana Reddy, these needs were translated into a product specification, which served as a strong basis for the development procedure. The translators, Chinmay J. Reddy and Syed Taahir Ahamed, have worked together to translate a wide range of user needs into tangible features and specifications that make the tool inclusive and applicable to a wide range of user demographics. The Heart Disease Predictor has the potential to have a significant influence on public health as wrap up this part of

the project. But it's important to recognize that this is only the start of our adventure. The tool's long-term success depends on validation, adaptability, and constant development

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